Computer Programming (CSN11101)
ASSIGNMENT-3

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Q1. Write a program to find if a year is leap year or not .

a) Use &&, || operators
b) Without using &&, || operators.

A1. (a)
CODE:

```
#include<stdio.h>

int main(){
  int n;
  printf("Enter the year : ");
  scanf("%d",&n);
  if(n%100==0 && n%400!=0){printf(" Not a Leap Year\n");}
  else if(n%4==0){printf("Leap Year\n");}
  else{printf("Not a leap year\n");}
}
```

OUTPUT:

```
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p1
Enter the year : 2024
Leap Year
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p1
Enter the year : 2025
Not a leap year
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p1
Enter the year : 2200
 Not a Leap Year
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p1
Enter the year : 2400
Leap Year
```

(b) CODE:

```
#include<stdio.h>

int main(){
  int n;
  printf("Enter the year : ");
  scanf("%d",&n);
  n%4!=0 ? printf("Not a leap year\n"):(
    n%100!=0 ? printf("Leap year\n"):(
      n%400!=0 ? printf("Not a leap year\n"):printf("Leap Year\n")
      )
  );
}
```

OUTPUT:

```
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p2
Enter the year : 2024
Leap year
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p2
Enter the year : 2025
Not a leap year
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p2
Enter the year : 2200
Not a leap year
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p2
Enter the year : 2400
Leap Year
```

Q2. Write a program to print the grade of a student according to the score. where A+ for 85 or above, A for 75 to 84, B+ for 65 to 74, B for 50 to 64, C for 30 to 49 and fail below 30.

A2. CODE:

```
#include<stdio.h>

int main(){
  int n;
  printf("Enter the marks : ");
  scanf("%d",&n);
  if(n>=85){printf("Grade : A+\n");}
  else if(n>=75){printf("Grade : A\n");}
  else if(n>=65){printf("Grade : B+\n");}
  else if(n>=50){printf("Grade : B\n");}
  else if(n>=30){printf("Grade : C\n");}
  else{printf("Fail\n");}
}
```

OUTPUT:

```
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p3
Enter the marks : 90
Grade : A+
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p3
Enter the marks : 80
Grade : A
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p3
Enter the marks : 70
Grade : B+
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p3
Enter the marks : 60
Grade : B
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p3
Enter the marks : 45
Grade : C
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p3
Enter the marks : 35
Grade : C
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p3
Enter the marks : 25
Fail
```

Q3. Write a program to reverse the digits of an integer.
A3. CODE:

```
#include<stdio.h>

int main(){
  int n;
  int answer=0;
  printf("Enter the number : ");
  scanf("%d",&n);
  while(n>=1){
    answer=answer*10+n%10;
    n=n/10;
  }
  printf("Reversed number : %d \n",answer);
}
```

OUTPUT:

```
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p4
Enter the number : 12344
Reversed number : 44321
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p4
Enter the number : 64728
Reversed number : 82746
```

Q4.Write a program to check if a given number is prime.
A4. CODE :

```
#include<stdio.h>

int main(){
  int n;
  int answer=0;
  printf("Enter the number : ");
  scanf("%d",&n);
  for(int i=2;i<n;i++){
    if(n%i==0){answer=1;break;}
  }
  if(answer){printf("The number is not prime\n");}
  else{printf("The number is prime\n");}
}
```

OUTPUT :

```
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p5
Enter the number : 17
The number is prime
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p5
Enter the number : 19
The number is prime
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p5
Enter the number : 15
The number is not prime
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p5
Enter the number : 2
The number is prime
```

Q5. Write a program to print all ASCII values and their equivalent characters.
A5. CODE :

```
#include<stdio.h>

int main(){
  for(char c=65;c<127;c++){
    printf("%d : %c \n",c,c);
  }
}
```

OUTPUT :

```
65 : A
66 : B
67 : C
68 : D
69 : E
70 : F
71 : G
72 : H
73 : I
74 : J
75 : K
76 : L
77 : M
78 : N
79 : O
80 : P
81 : Q
82 : R
83 : S
84 : T
85 : U
86 : V
87 : W
88 : X
89 : Y
90 : Z
```

```
91 : [
92 : \
93 : ]
94 : ^
95 : _
96 : `
97 : a
98 : b
99 : c
100 : d
101 : e
102 : f
103 : g
104 : h
105 : i
106 : j
107 : k
108 : l
109 : m
110 : n
111 : o
112 : p
113 : q
114 : r
115 : s
```

```
115 : s
116 : t
117 : u
118 : v
119 : w
120 : x
121 : y
122 : z
123 : {
124 : |
125 : }
126 : ~
```

Q6.Write a program for simple calculator using switch case in C.
A6. CODE:

```
#include<stdio.h>

int main(){
  int n;
  printf("1 : Addition\n");
  printf("2 : Subtraction\n");
  printf("3 : Product\n");
  printf("4 : Division\n");
  printf("Enter the number : ");
  scanf("%d",&n);
  switch(n){
    case(1):
      int a,b;
      printf("Enter the first value : ");
      scanf("%d",&a);
      printf("Enter the second value : ");
      scanf("%d",&b);
      printf("The addition is : %d\n",a+b);
      break;
    case(2):
      int c,d;
      printf("Enter the first value : ");
      scanf("%d",&c);
      printf("Enter the second value : ");
      scanf("%d",&d);
      printf("The subtraction is : %d\n",c-d);
      break;
    case(3):
      int e,f;
      printf("Enter the first value : ");
      scanf("%d",&e);
      printf("Enter the second value : ");
      scanf("%d",&f);
      printf("The product is : %d\n",e*f);
      break;
    case(4):
      int g,h;
      printf("Enter the first value : ");
      scanf("%d",&g);
      printf("Enter the second value : ");
      scanf("%d",&h);
      printf("The division is : %d\n",g/h);
      break;
  }
}
```

OUTPUT:

```
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p7
1 : Addition
2 : Subtraction
3 : Product
4 : Division
Enter the number : 1
Enter the first value : 12
Enter the second value : 13
The addition is : 25
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p7
1 : Addition
2 : Subtraction
3 : Product
4 : Division
Enter the number : 2
Enter the first value : 15
Enter the second value : 6
The subtraction is : 9
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p7
1 : Addition
2 : Subtraction
3 : Product
4 : Division
Enter the number : 3
Enter the first value : 5
Enter the second value : 4
The product is : 20
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p7
1 : Addition
2 : Subtraction
3 : Product
4 : Division
Enter the number : 4
Enter the first value : 36
Enter the second value : 4
The division is : 9
```

Q7.Write a program to find the factorial value of any number entered through the keyboard.

CODE:

```
#include<stdio.h>

int main(){
  int n=0;
  printf("Enter the number : ");
  scanf("%d",&n);
  int answer=1;
  for(int i=1;i<=n;i++){
    answer*=i;
  }
  printf("The factorial is : %d \n",answer);
}
```

OUTPUT:

```
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p8
Enter the number : 5
The factorial is : 120
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment 3$ ./p8
Enter the number : 4
The factorial is : 24
```